

Stochastics

(3. Stochastic Calculus)

(The calculus of W)

Variations

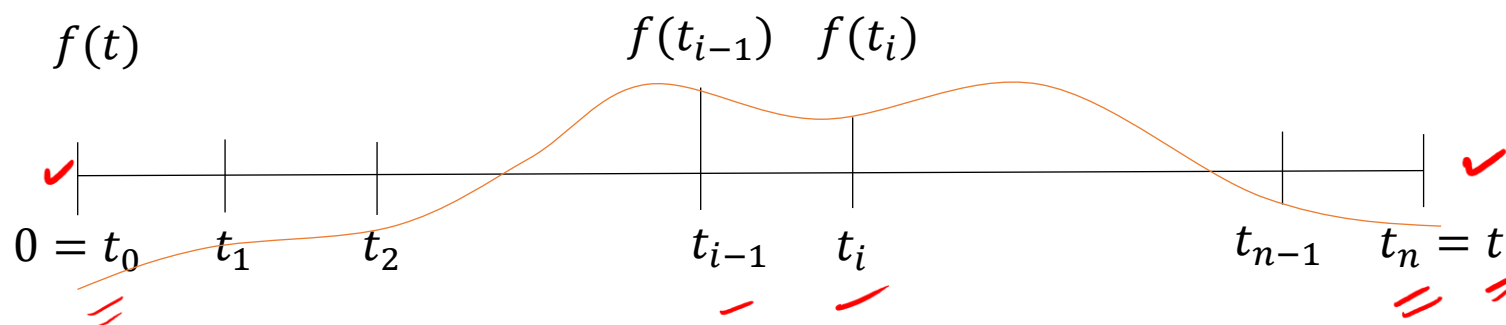
- Consider a time interval $(0, t)$. Let's break down this interval into n sub intervals $0 = t_0 < t_1 < t_2 < \dots < t_{n-1} < t_n = t$. **Total Variation (TV)** of a function $f(t)$ is defined as

$$\text{TV} = \lim_{n \rightarrow \infty} \sum_{i=1}^n |f(t_i) - f(t_{i-1})|$$

- **Quadratic Variation (QV)** of a function $f(t)$ is defined as

$$\text{QV} = \lim_{n \rightarrow \infty} \sum_{i=1}^n |f(t_i) - f(t_{i-1})|^2$$

$\Delta t = t/n \rightarrow 0$
 Δt



Variations

► Let's analyze three cases of smooth functions in the interval $0 \leq x \leq 2$ and calculate TV and QV

□ $f(x) = x, 0 \leq x \leq 2$ ✓ $f'(x) = 1$ ✓ $f'(x) > 0$
 $f'(x) < 0$

□ $f(x) = x^2, 0 \leq x \leq 2$ ✓ $f'(x) = 2x$ ✓

□ $f(x) = 3x^2 - 5x, 0 \leq x \leq 2$, $f'(x) = 6x - 5 = 0$
 $x^* = 5/6$

⊕ $f'(x) =$
 $(dx)^2, (dt)^2$
 $\rightarrow 0$

⊕ TV = $\int_0^2 \underline{f'(x)} dx = \int_0^2 1 \cdot dx = \textcircled{2}$
 $\int_0^2 2x dx = [x^2]_0^2 = 4$ ✗

Variation of smooth functions

- ▶ For a **monotonic function** such as $f(x) = x, 0 \leq x \leq 2$ or $f(x) = x^2, 0 \leq x \leq 2$ with increasing number of steps
 - ▶ the total variation (**TV**) remains **constant**
 - ▶ the quadratic variation (**QV**) approaches **0**.

- ▶ For a **non-monotonic function** such as $f(x) = 3x^2 - 5x, 0 \leq x \leq 2$ where the slope changes sign, with increasing number of steps
 - ▶ the total variation (**TV**) will oscillate but eventually converge to a value and
 - ▶ the quadratic variation (**QV**) approaches **0**.

Exercise – Show that the TV for the function $f(x) = 3x^2 - 5x, 0 \leq x \leq 2$ converges to $\frac{37}{6}$ as $n \rightarrow \infty$

HW

$\frac{37}{6}$
= 1.66...

Variations of Brownian Motion

- ▶ For **Brownian motion** $\{\underline{W}_t, t \geq \underline{0}\}$, with increasing number of steps
 - ▶ the total variation (**TV**) explodes (reaches infinity !)
 - ▶ the quadratic variation (**QV**) approaches **t**.

$$TV = \lim_{n \rightarrow \infty} \sum_{i=1}^n |W(t_i) - W(t_{i-1})| = \infty$$

$$QV = \lim_{n \rightarrow \infty} \sum_{i=1}^n |W(t_i) - W(t_{i-1})|^2 = t$$

Know your W and dW

► Important results to remember

$$dW_t \sim N(0, dt)$$

$$dW_t = Z\sqrt{dt}$$

$$\mathbb{E}[dW_t] = 0$$

$$\mathbb{V}[dW_t] = dt$$

$$\mathbb{E}[dW_t^2] = dt$$

$$\mathbb{V}[dW_t^2] = 2 \times dt^2 \rightarrow 0$$

$$\mathbb{E}[dW_t \cdot dt] = 0$$

$$\mathbb{V}[dW_t \cdot dt] = dt^3 \rightarrow 0$$

$$W_t \sim N(0, t)$$

$$W_t = Z\sqrt{t}$$

$$\mathbb{E}[W_t] = 0$$

$$\mathbb{V}[W_t] = t$$

$$\mathbb{E}[W_t^2] = t$$

$$\mathbb{V}[W_t^2] = 2t^2$$

$$\mathbb{E}[W_t^3] = 0$$

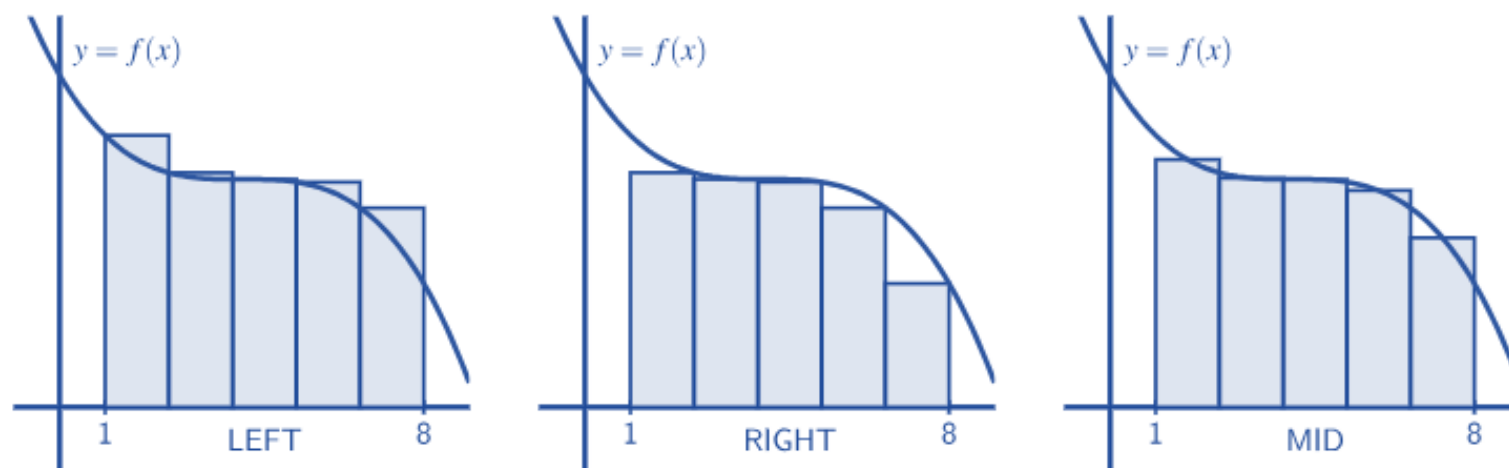
$$\mathbb{E}[W_t^4] = 3t^2$$

Stochastic Integral (I)

► An integration w.r.t **Brownian** motion is called **Stochastic Integral**.

$$I_t = \int_0^t f(s, W_s) dW_s$$

There is a crucial difference between ordinary integral calculus and stochastic calculus. In ordinary integral calculus, the **left-rule**, **right-rule** and **mid-point** rule result converge to the true integral as $n \rightarrow \infty$. It's not the case with stochastic calculus.



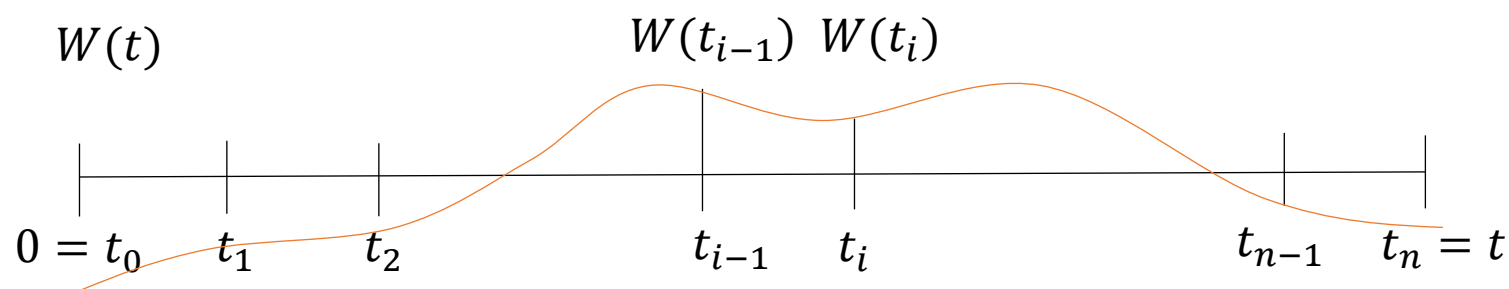
Stochastic Integral (II)

- In stochastic integral, the different rules do not converge to the same value. For example, assume n sub intervals $0 = t_0 < t_1 < t_2 < \dots < t_{n-1} < t_n = t$ between $[0, t]$.

$$I_t = \int_0^t W_s dW_s = \lim_{n \rightarrow \infty} \sum_{i=1}^n W_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) = \frac{1}{2} (W_t^2 - t) \quad (\text{Ito})$$

$$I_t = \int_0^t W_s dW_s = \lim_{n \rightarrow \infty} \sum_{i=1}^n W_{t_i} (W_{t_i} - W_{t_{i-1}}) = \frac{1}{2} (W_t^2 + t)$$

$$I_t = \int_0^t W_s dW_s = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{W_{t_{i-1}} + W_{t_i}}{2} \right) (W_{t_i} - W_{t_{i-1}}) = \frac{1}{2} W_t^2 \quad (\text{Stratonovich})$$



Ito Integral



▶ We now formally define **Ito integral** as

$$I_t = \int_0^t f(s, W_s) dW_s = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_{i-1}, W_{t_{i-1}})(W_{t_i} - W_{t_{i-1}})$$



- ✓ Left Point
- ✓ Independent increments
- ✓ New Differentiation rules
- ✓ Martingales

Ito Integral – Constant functions

- Let's evaluate the following Ito integrals. The approach is to discretize the integration into a sum and use the property that Brownian increments are independent and gaussian. Let's start with some constant functions

$$\int_0^t dW_s = W_t$$

$$\int_0^t 2 \cdot dW_s = 2W_t$$

Ito Integral – Step functions

- We can proceed with the same approach for a step function $f(s)$. Integrating a step function is straightforward. We just need to break the interval based on the discontinuity of the function

$$\int_0^t f(s) dW_s$$

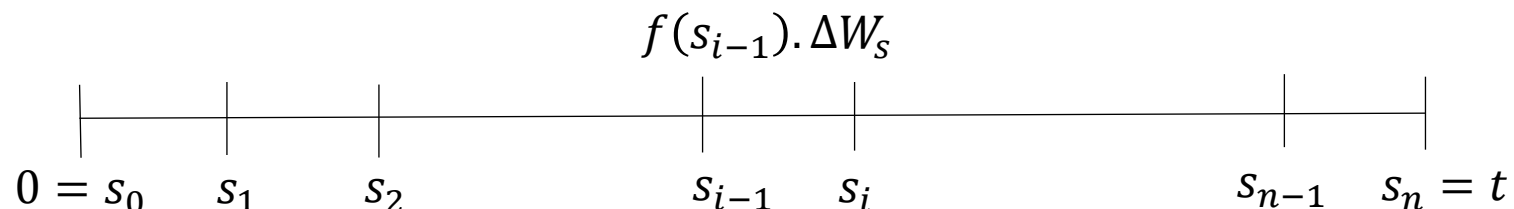
Exercise – Evaluate the integral $\int_1^3 f(s) dW_s$ where,

$$f(s) = \begin{cases} 1, & 1 \leq s < 2 \\ 2, & 2 \leq s < 3 \\ 0, & \text{otherwise} \end{cases}$$

Ito Integral – continuous deterministic functions

► For any deterministic function $f(s)$, we can proceed as per the **Ito integral** definition

$$\int_a^b f(s) dW_s = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(s_{i-1}) \Delta W_s$$



The distribution of the integral is $\int_a^b f(s) dW_s \sim \lim_{n \rightarrow \infty} \sum_{i=1}^n N\left(0, f(s_{i-1})^2 (s_i - s_{i-1})\right)$

$$\sim N\left(0, \underbrace{\int_a^b f^2(s) ds}_{\text{(ordinary integral)}}\right)$$

Examples



► Integrate the following

$$\int_0^t (a + bs) dW_s$$

$$\int_0^t A e^{-Bs} dW_s$$

Ito Integral is martingale

► Ito integrals are **martingales**. In our previous example define the stochastic process I_t as

$$I_t = \int_a^t f(s) dW_s$$

Then,

$$\mathbb{E}[I_t | \mathcal{F}_u] = I_u$$

Exercise – Show that Ito integral for functions of time and Wiener process $f(t, W_t)$ is also martingale using similar arguments.

Properties of Ito Integrals

► Ito integrals have the following properties. Define I_t as

$$I_t = \int_0^t f(s, W_s) dW_s$$

Then, 1) I_t is a **martingale**

2) $\mathbb{E}[I_t] = \mathbf{0}$

(Ito isometry) 3) $\mathbb{E} \left[\left(\int_0^t f(s, W_s) dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^t f^2(s, W_s) ds \right] = \int_0^t \mathbb{E}[f^2(s, W_s)] ds$

4) Sample paths of I_t are **continuous**.

► The distribution of Ito integral of a deterministic function is Normal. But the distribution of Ito integral of a **stochastic** function **does not have to be normal**.